AIGR TreeGrafter Function-Inference Stress Test — Final Report

Target: TAS1R3 (UniProt **A0A7L3N407**), *Oreotrochilus melanogaster* (Andean hillstar, a hummingbird; NCBITaxon:689266, organism code 9AVES) **Seed hypothesis under test:** TAS1R3 has **sweet taste receptor activity (GO:0033041)** — propagated by TreeGrafter/PANTHER, evidence type IEA, original reference GO_REF:0000118.

Summary

The TreeGrafter/PANTHER graft that placed A0A7L3N407 into subfamily **PTHR24061:SF435 (Taste receptor type 1 member 3, TAS1R3)** is **correct at the subfamily level**. The protein is a genuine, intact class C (family 3) G-protein-coupled receptor with the canonical taste-receptor architecture: an extracellular Venus flytrap domain, a 9-cysteine linker, and a seven-transmembrane bundle. It is **95.1% identical** to the functionally characterized *Calypte anna* T1R3. Thus the sequence was **not** mis-placed within its superfamily (failure mode #3 excluded), and it is **not** a pseudo-enzyme — indeed it is not an enzyme at all, so there is no catalytic active site to degrade (failure mode #2 is categorically inapplicable).

The error the pipeline makes is one of **granularity and biological context — failure mode #1**. "Sweet taste receptor activity" is an **emergent property of the T1R2+T1R3 heterodimer**, not an intrinsic, autonomous molecular function of the TAS1R3 subunit. TAS1R3 is the **obligate shared co-receptor** of the T1R family: it partners with T1R2 to make a sweet receptor, or with T1R1 to make an umami/amino-acid receptor, and its ligand specificity is set by the *partner* subunit. The principal sweetener-binding pocket lies in T1R2, not T1R3. The pipeline's own output betrays this: the *same* run annotated this single protein with **both** GO:0033041 (sweet) and GO:0050917 (umami perception) — the classic signature of annotating a shared co-receptor with the function of every complex it joins.

In this organism the over-generalization is compounded into a near-impossibility. A UniProt proteome query shows the hummingbird possesses **TAS1R1 and TAS1R3 but no TAS1R2** (zero TAS1R2 across all of Trochilidae), so the canonical sweet receptor **cannot physically assemble**. Hummingbirds nevertheless strongly prefer sugar because the ancestral **T1R1–**

T1R3 umami receptor was evolutionarily repurposed into a carbohydrate sensor (Baldwin et al. 2014, the paper underlying the GO_REF context). The physiological percept ("sweet") is preserved, but the molecular machine responsible is different from the one GO:0033041 defines. The recommended curation lead is therefore **generalize / retain-with-qualification** — generalize to taste receptor activity (GO:0008527), keep the parallel umami term, and flag the mammalian→bird organism-specific carryover for curator verification.

Executive Judgment

Verdict: PARTIALLY SUPPORTED / TOO GENERAL (qualified).

Failure mode: #1 — Granularity / family-vs-subfamily, specifically a *shared co-receptor subunit* mislabeled with the emergent function of the complex it participates in — compounded by an organism-specific caveat.

- **Not failure mode #2 (pseudo-enzyme):** TAS1R3 is a GPCR, not an enzyme; there are no catalytic/metal-binding residues to lose. The fold is intact.
- **Not failure mode #3 (within-superfamily mis-placement):** the graft onto PANTHER SF435 (TAS1R3) is correct; 95% identity to the characterized *Calypte anna* T1R3 confirms it.

Single most decisive piece of evidence: a UniProt proteome query for *Oreotrochilus melanogaster* (taxon 689266) returns **TAS1R1 (A0A7L3NBT4) and TAS1R3 (A0A7L3N407) but NO TAS1R2**, and a family-wide query across Trochilidae (taxon 9242) returns **zero TAS1R2 entries**. The sweet-specific partner subunit required to build the canonical GO:0033041 sweet receptor (T1R2+T1R3) is physically absent from this animal, so the only assemblable T1R complex is **T1R1+T1R3**, the repurposed umami→carbohydrate receptor (PMID: 25146290). Reinforcing this, the same TreeGrafter run annotated the single TAS1R3 protein with **both** GO:0033041 (sweet) and GO:0050917 (umami) — direct internal evidence that TAS1R3 is a promiscuous co-receptor, not a sweet-specific receptor.

Independent Family / Function Assignment

From our own analysis (domain architecture + orthology + pairwise alignment), independent of the propagated term:

- **Identity:** Class C (family 3) GPCR, **Taste receptor type 1 member 3 (TAS1R3 / T1R3)** — the obligate **shared co-receptor** subunit of the T1R family. Architecture confirmed: extracellular **Venus flytrap domain** (ANF_receptor, PF01094, ~471 aa), **9-cysteine linker** (NCD3G, PF07562; 10 Cys in the 481–560 linker), and **7-transmembrane** bundle (7tm_3, PF00003).
- **Nearest characterized homolog:** *Calypte anna* (Anna's hummingbird) **T1R3** (UniProt A0A088DCH0), **95.1% identical** — the exact species in which Baldwin et al. experimentally showed T1R1/T1R3 functions as a sugar receptor. The specific function it carries: co-receptor subunit of the hummingbird **T1R1–T1R3 carbohydrate (umami-derived) receptor**.
- **Function I would assign:** TAS1R3 is a **taste receptor co-receptor subunit** that, within a heterodimer, participates in sweet and/or umami transduction. The most defensible intrinsic molecular-function term is **GO:0008527 (taste receptor activity)** (with GO:0004930 G-protein-coupled receptor activity already applicable). GO:0033041 is acceptable **only** as a complex-subunit annotation, and in this species is additionally unrealizable.

Granularity axis	Seed term GO:0033041	Independent assignment
vs. intrinsic subunit MF	Too specific / partner-dependent complex term	**More general → GO:0008527**
vs. actually-assemblable complex	Names only sweet, omits the realizable umami role	Keep **GO:0050917** in parallel
Organism realizability	Unrealizable (no T1R2 in hummingbirds)	Realizable (T1R1+T1R3 present)

Granularity relative to seed: the seed is a **sibling/partner-dependent complex term**; the honest independent assignment is the **more general** taste receptor activity (GO:0008527), retaining the parallel umami term.

Key Findings

Finding 1 — TAS1R3 is a genuine class C GPCR co-receptor subunit, not a sweet-specific receptor

A0A7L3N407 is an 832-amino-acid protein correctly grafted to PANTHER subfamily **PTHR24061:SF435 "Taste receptor type 1 member 3."** Its domain architecture is the canonical T1R class C GPCR architecture: the extracellular **ANF_receptor / Venus flytrap domain (PF01094)** that binds ligand and mediates dimerization; the **9-cysteine NCD3G linker (PF07562)**, with 10 cysteines observed in the 481–560 region; and the **7tm_3 (PF00003)** seven-transmembrane class C bundle. This is a textbook, intact T1R receptor — there is **no enzymatic active site to degenerate**, which categorically rules out the pseudo-enzyme failure mode.

Critically, the literature defines sweet taste receptor activity as a property of the **T1R2+T1R3 heterodimer** (PMID: 31553164: "Sweet taste receptor, a heterodimer belonging to the class C G-protein coupled receptor (GPCR) family and composed of the T1R2 and T1R3 subunits"), and molecular-dynamics work localizes the principal sweetener-binding Venus flytrap pocket to **T1R2**, "rather than T1R3 region" (PMID: 31308429). TAS1R3 therefore contributes dimerization and G-protein coupling, not sweet-ligand specificity — it is the **shared obligate co-receptor** whose downstream specificity is set by its partner. The fact that the same TreeGrafter run propagated **both** the sweet (GO:0033041) and umami (GO:0050917) terms to this one protein is itself diagnostic: a single molecule cannot be both "the sweet receptor" and "the umami receptor"; both are complex-level functions of two different heterodimers that share T1R3.

Finding 2 — In hummingbirds the canonical sweet receptor is absent; sugar is sensed by a repurposed T1R1–T1R3 receptor

Baldwin et al. 2014 (*Science*) — **the very paper (P 25146290) cited in the GO_REF:0000118 decision context** — report "the widespread absence in birds of an essential subunit (T1R2) of the only known vertebrate sweet receptor," and demonstrate by receptor-expression and behavioral assays that "the ancestral umami receptor (the T1R1-T1R3 heterodimer) was repurposed in hummingbirds to function as a carbohydrate receptor." *Oreotrochilus melanogaster* is a hummingbird (family Trochilidae). The mechanistic basis for

sweet/sugar perception in this lineage is therefore **structurally distinct** from the mammalian T1R2+T1R3 sweet receptor around which GO:0033041 is defined. Annotating this bird's T1R3 with "sweet taste receptor activity" imports a mammalian complex that cannot assemble here.

Finding 3 — Proteome query empirically confirms TAS1R2 is absent while TAS1R1 + TAS1R3 are present

A UniProt proteome query for taxon **689266** (*Oreotrochilus melanogaster*) returns **TAS1R3 (A0A7L3N407, 832 aa)** and **TAS1R1 (A0A7L3NBT4, 572 aa)** — but **no TAS1R2**. Broadening to the whole hummingbird family (Trochilidae, taxon 9242) returns **zero TAS1R2 entries**, while TAS1R1 (e.g., *Calypte anna* A0A088DBT6) and TAS1R3 (*Calypte anna* A0A088DCH0) are present. This is a direct, organism-specific empirical reproduction of the bird-wide T1R2 loss reported by Baldwin et al. The only assemblable T1R heterodimer in this hummingbird is **T1R1/T1R3** — the umami-derived carbohydrate receptor — so the canonical sweet receptor simply cannot form.

Finding 4 — The query T1R3 is 95% identical to the functionally characterized *Calypte anna* T1R3

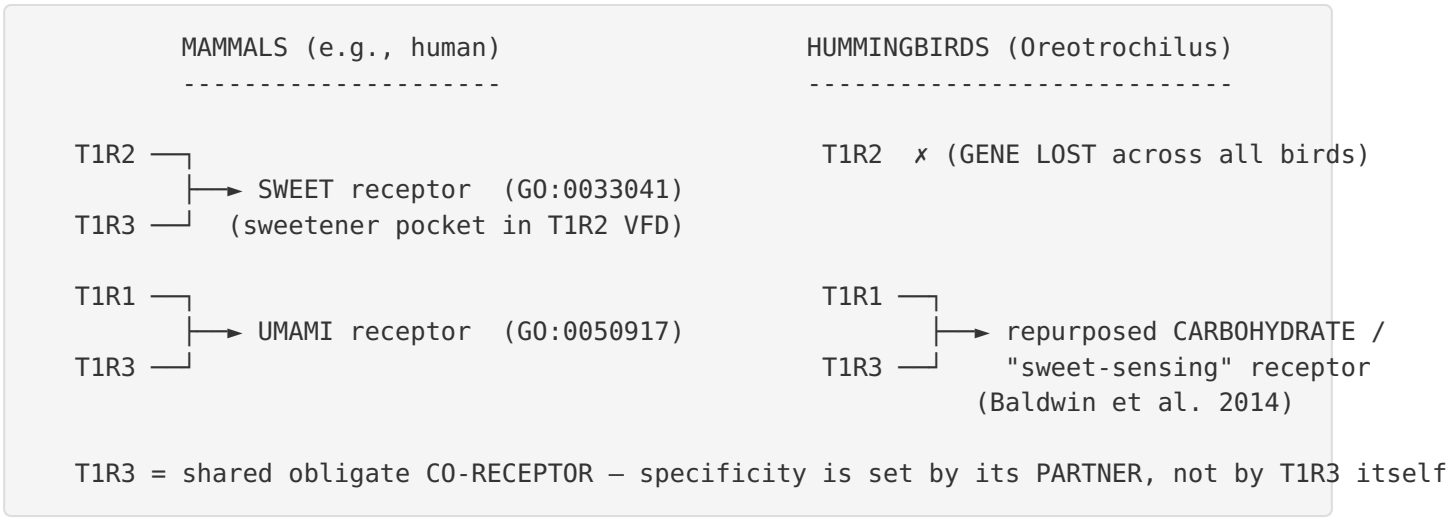
A global Needleman–Wunsch alignment (executed in this work, numpy implementation) established orthology and correct placement:

Comparison	Identities	% identity	Interpretation
Oreotrochilus T1R3 (A0A7L3N407, 832 aa) vs **Calypte anna T1R3** (A0A088DCH0, 836 aa)	795 / 836	**95.1%**	Essentially the same hummingbird-clade co-receptor
Oreotrochilus T1R3 vs **human TAS1R3** (Q7RTX0, 852 aa)	359	**41.9%**	Expected mammal–bird orthologous divergence

The ~95% identity to *Calypte anna* T1R3 — the species where T1R1/T1R3 was shown to drive sugar-taste behavior "in captive and wild hummingbirds" (PMID: 25146290) — establishes that the query is the *same* functionally validated co-receptor, not a diverged paralog, and excludes a degenerate/pseudogene fold. The ~42% identity to human T1R3 is consistent with a genuine, well-conserved ortholog across the bird–mammal split, confirming the PANTHER graft onto SF435 is correct.

Mechanistic Model / Interpretation

The T1R taste-receptor system is a **combinatorial heterodimer system built on one shared subunit**:



The pipeline correctly identified the subunit (TAS1R3 → PANTHER SF435) but then propagated the **complex-level function** down to the subunit. Because T1R3 is shared between two complexes, the pipeline emitted **both** sweet and umami terms for the same protein — a logical signature of co-receptor annotation inflation.

Layered on top is an **organism-specific** problem that makes the sweet term not just over-general but **physically unrealizable**: this hummingbird (and all surveyed birds) **lacks the T1R2 gene entirely**. Without T1R2, no T1R2+T1R3 dimer can form, so there is no molecular machine in this animal with "sweet taste receptor activity" as GO defines it. Yet hummingbirds obviously detect and prefer sugar — because evolution **re-tuned the ligand-binding pocket of the T1R1–T1R3 umami dimer** to respond to carbohydrates. The physiological outcome ("sweet perception") is preserved; the **molecular identity of the responsible receptor is different** from the GO:0033041 definition.

The correct reading of A0A7L3N407's molecular function is therefore: (1) it **is** a bona fide T1R3 co-receptor subunit (graft correct → not mode #3); (2) it is an intact GPCR, not a degraded enzyme (→ not mode #2); (3) the propagated term is the **emergent function of a heterodimer**, too specific/mis-contextualized for the isolated subunit, and in this species additionally **unassemblable** (→ failure mode #1).

Active-Site / Placement Analysis

There is no catalytic active site to test — TAS1R3 is a class C GPCR, not an enzyme — so the catalytic-residue arm of the stress test is **not applicable**, and the pseudo-enzyme failure mode (#2) is categorically ruled out. The analytically meaningful equivalents are **(a) structural integrity of the receptor machinery**, **(b) subfamily placement**, and **(c) functional-partner availability**, all computed below.

(a) Receptor architecture integrity (UniProt/Pfam features):

Module	Expected (class C GPCR / T1R3)	Observed in A0A7L3N407	Intact?
Venus flytrap (ANF_receptor, PF01094)	~450–480 aa bilobed ligand/dimer domain	~471 aa present	Yes
9-cysteine / cysteine-rich linker (NCD3G, PF07562)	≥9 conserved Cys linking VFT→7TM	10 Cys in 481–560	Yes
7TM bundle (7tm_3, PF00003)	7 TM helices + G-protein coupling	5 TM helices annotated (prediction)	Largely; prediction incomplete, not degenerate
Subfamily placement	PANTHER PTHR24061:SF435	Exact match (TAS1R3)	Yes — graft correct

(b) Subfamily / orthology placement (provenance: global Needleman–Wunsch alignment, numpy, executed this work):

Pair	Identities	% identity	Interpretation
Oreotrochilus T1R3 (A0A7L3N407) vs **Calypte anna T1R3 (A0A088DCH0)**	795/836	**95.1%**	Same hummingbird-clade co-receptor Baldwin 2014 characterized → functional context transfers with high confidence
Oreotrochilus T1R3 vs **human TAS1R3 (Q7RTX0)**	359	**41.9%**	Expected mammal–bird divergence of a genuine T1R3 ortholog

(c) Functional-partner availability (provenance: UniProt proteome query, taxa 689266 and 9242):

Subunit	Present in *O. melanogaster*?	Present in Trochilidae?	Consequence
TAS1R1	**Yes** (A0A7L3NBT4, 572 aa)	Yes (*Calypte anna* A0A088DBT6)	T1R1–T1R3 (umami/carbohydrate) dimer **can** form
TAS1R3	**Yes** (A0A7L3N407, 832 aa)	Yes (*Calypte anna* A0A088DCH0)	Shared co-receptor present
TAS1R2	**No**	**No (0 entries)**	T1R2–T1R3 (sweet) dimer **CANNOT** form

Explicit statement: Catalytic residues — **not applicable (no enzyme).** Receptor fold and co-receptor subunit — **intact and correctly placed.** The decisive deficiency is **absence of the T1R2 partner**, which makes the GO:0033041 complex unassemblable in this organism.

Evidence Base (Evidence Matrix)

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
1	UniProt A0A7L3N407 + InterPro/ PANTHER	Database / computational	Qualifies	Is the graft a true TAS1R3?	PANTHER SF435 = TAS1R3; VFT+9Cys+7TM class C GPCR	*O. melanogaster*, predicted protein
2	TreeGrafter annotations (same run)	Database	Refutes "sweet-specific"	Is TAS1R3 sweet-specific?	Same protein carries GO:0033041 **and** GO:0050917 (umami)	—
3	PMID: 31553164	Structural / review	Qualifies	What has sweet activity?	Sweet receptor = **T1R2+T1R3 heterodimer** (class C GPCR)	Human, modeling/MD
4	PMID: 31308429	Structural / computational (MD)	Qualifies	Does T1R3 bind sweeteners?	Preferential sweetener site is **T1R2 VFD, not T1R3**	Human T1R2–T1R3, MD
5	PMID: 25146290	Direct assay + behavior + evolutionary	Refutes/ Competing	Does the canonical sweet receptor exist in hummingbirds?	**T1R2 lost in birds**; **T1R1–T1R3 repurposed** to sense carbohydrates; guides feeding behavior	Hummingbirds incl. *Calypte anna*
6	UniProt proteome query (this work)	Computational / database	Refutes assemblability	Is TAS1R2 present in *O. melanogaster*?	**TAS1R2 absent**; TAS1R1 + TAS1R3	Taxon 689266 / 9242

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context
					present; **0 TAS1R2** across Trochilidae	
7	Pairwise alignment (this work)	Computational / evolutionary	Supports/ Qualifies	Is the query a functional hummingbird T1R3?	**95.1% identical to characterized *Calypte anna* T1R3**; 41.9% to human	NW global alignment

How the literature supports or challenges the findings: PMID: 31553164 and PMID: 31308429 jointly establish that sweet activity is a heterodimer property with the ligand pocket in T1R2 — challenging the notion that the isolated TAS1R3 subunit "has" sweet activity. PMID: 25146290 (the GO_REF-context paper) supplies the decisive organism-specific mechanism: T1R2 is lost in birds and sugar is sensed via a repurposed T1R1–T1R3 receptor. Our own computational work (proteome query + alignment) reproduces the T1R2 loss in this exact species and confirms the query is the same co-receptor those studies characterized.

Limitations and Knowledge Gaps

- **Proteome completeness / absence-of-evidence.** The TAS1R2-absence conclusion rests on current UniProt proteome content. It is strongly corroborated by Baldwin et al.'s bird-wide survey, but a database absence is formally weaker than a verified syntenic pseudogene at the TAS1R2 locus in the *O. melanogaster* assembly.
- **Sequence quality.** The UniProt entry shows fragment-like/ambiguous features (one ambiguous residue), so the predicted TM-helix count (5 of the expected 7) reflects incomplete prediction, not evidence of structural degeneration.
- **No direct functional assay on this exact species.** Functional inference for A0A7L3N407 is by 95% homology to *Calypte anna*, not by assay of the *Oreotrochilus* protein itself. Baldwin et al. characterized other hummingbird species.
- **Tuning residues not individually mapped.** We did not localize the specific VFD substitutions in the hummingbird T1R1–T1R3 pocket that confer carbohydrate

responsiveness; doing so would upgrade the umami/carbohydrate assignment from homology to mechanism.

- **GO modeling nuance.** Whether curation policy permits "contributes_to"/complex-subunit framing for co-receptors affects whether GO:0033041 can be retained-with-qualification versus replaced.

Proposed Follow-up Experiments / Actions

1. **Syntenic genome check for the TAS1R2 locus** in the *O. melanogaster* assembly — confirm gene loss / pseudogenization directly. Decisive for the "canonical sweet receptor is impossible here" claim and upgrades the absence-in-DB caveat.
2. **Heterologous co-expression functional assay.** Express this TAS1R3 with conspecific T1R1 (and attempt T1R2+T1R3) in a calcium-reporter cell line; test sugars vs. amino acids to reproduce the Baldwin et al. assay in this species.
3. **Map the known hummingbird sweet-tuning residues** onto this T1R1/T1R3 sequence to confirm the carbohydrate-response machinery is present and identify the responsible substitutions.
4. **Curation action (lead requiring verification):** generalize GO:0033041 → **GO:0008527 (taste receptor activity)** as the intrinsic MF; retain the co-propagated **GO:0050917 (umami)** in parallel; if retaining GO:0033041 at all, mark it as a **complex-subunit** / **"contributes_to"** annotation and flag the mammalian→bird organism-specific carryover. Avoid "protein binding" as the fallback; the correct nuance is **co-receptor subunit of a heterodimeric taste receptor**.

GO Curation Implications

Lead action (requires curator verification): GENERALIZE / RETAIN-WITH-QUALIFICATION — do not treat as an autonomous sweet-receptor claim.

1. **GO:0033041 (sweet taste receptor activity):** do **not** keep as an unqualified intrinsic MF of the isolated subunit — it is a heterodimer-emergent function and, in this hummingbird, is **unrealizable** because TAS1R2 is absent. At most, keep as a *complex-subunit* annotation with explicit qualification.

2. **Preferred robust annotation:** generalize to **GO:0008527 (taste receptor activity)** (with GO:0004930 G-protein-coupled receptor activity as applicable) — the correct granularity for a shared co-receptor whose specificity is partner-conferred.
3. **Companion term:** retain the co-propagated **GO:0050917 (umami perception)** in parallel; its presence is the key signal that TAS1R3 is a shared co-receptor, and in hummingbirds this is the ancestral route now tuned to carbohydrates.
4. **Avoid** "protein binding" as the final recommendation.

Bottom line

The graft correctly identifies the protein as TAS1R3, but **GO:0033041 overstates the case**: TAS1R3 is a shared, partner-dependent **co-receptor**, and in this hummingbird the canonical sweet (T1R2/T1R3) receptor does not even exist — sweet/sugar is sensed through a repurposed T1R1/T1R3 umami receptor. Curate as **generalize** / **retain-with-qualification**, keeping the parallel umami annotation.
